Result:

Any comparision sort will take $\Omega(n \log n)$ time.

(↑ lower bound)

Overcoming Strategies:

1. Do not use comparision
2. Put some conditions on the input.

Sorting in Linear Time :

(without using comparision).

Counting Sort :

Input: $A[1, \ldots, n]$ wherein each

$A[i] \in \{1, \ldots, k\}$ $k \in \mathbb{N}$.

Output: $B[1, \ldots, n]$

Auxillary Storage: $C[0, \ldots, k]$

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Counting Sort (A, B, C):

1. For $i \leftarrow 0$ to k — $\Theta(k)$
2. $\quad C[i] \leftarrow 0$
3. For $j \leftarrow 1$ to n — $\Theta(n)$
4. $\quad C[A[j]] \leftarrow C[A[j]] + 1$
5. For $i \leftarrow 1$ to k — $\Theta(k)$
6. $\quad C[i] \leftarrow C[i] + C[i+1]$
7. For $j \leftarrow n$ down to 1 — $\Theta(n)$

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8. $B[C[A[j]]] \leftarrow A[j]$

9 $C[A[j]] \leftarrow C[A[j]] - 1$.

Analysis:

$$\Theta(k) + \Theta(n) + \Theta(k) + \Theta(n)$$

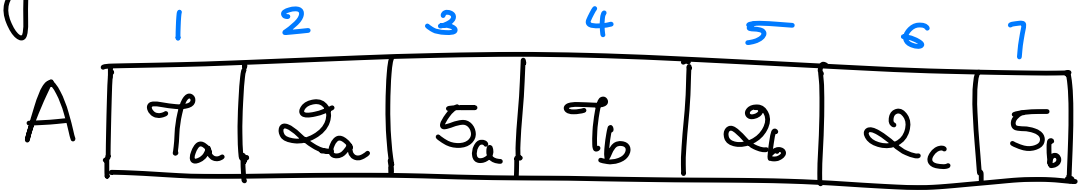
$$= \Theta(n+k).$$

The algorithm takes $\Theta(n+k)$ time.

It is linear if $k \in \Theta(n)$.

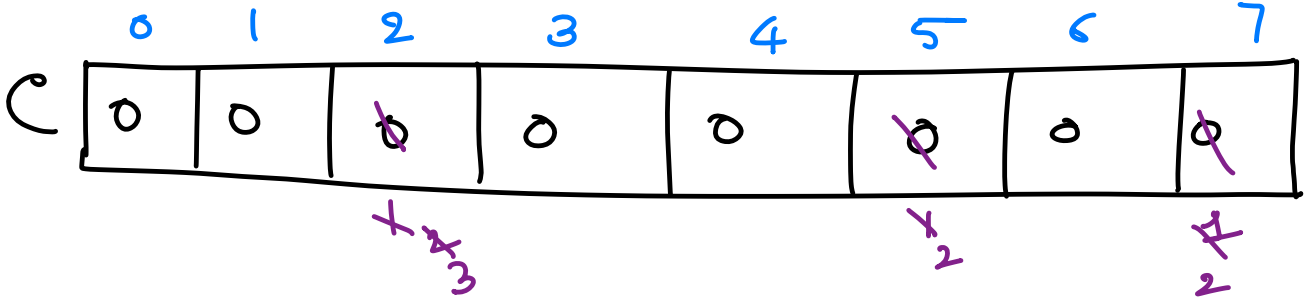
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eg:

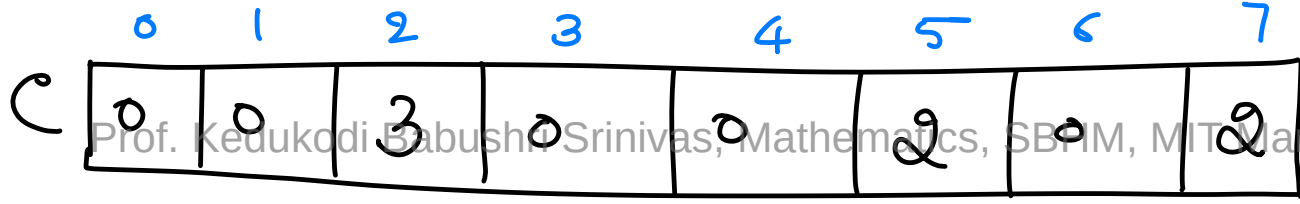


$k = 7$.

Step1: Initializing the Auxillary array



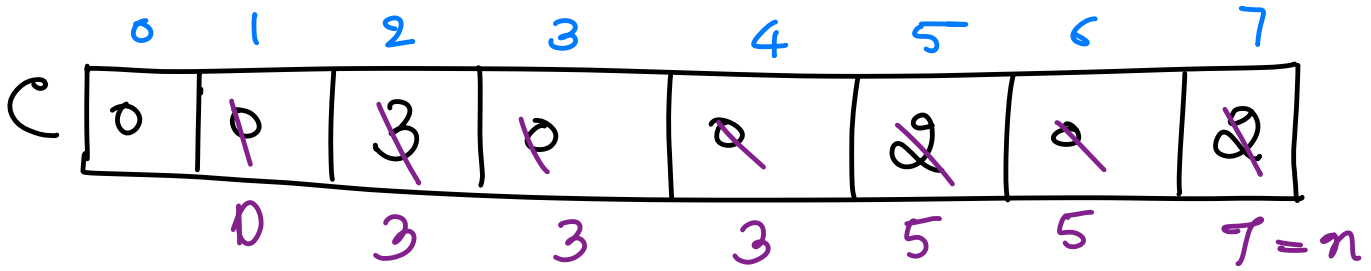
Step2: Filling the Auxillary Array:

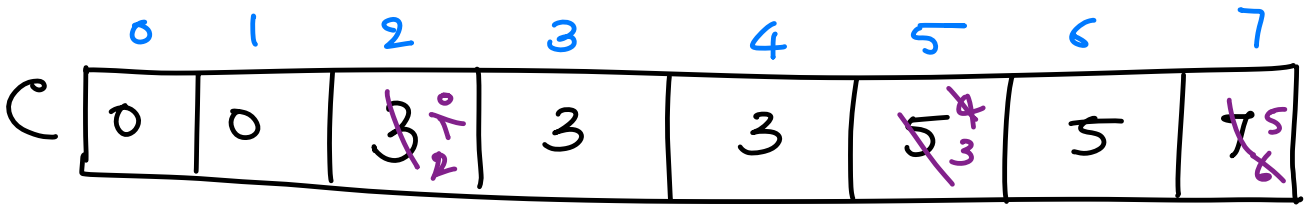


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In this step, it counts the number of occurences of the number i $0 \le i \le k$ in Array A. (frequency)

Step 3: (Cummulative Distribution).

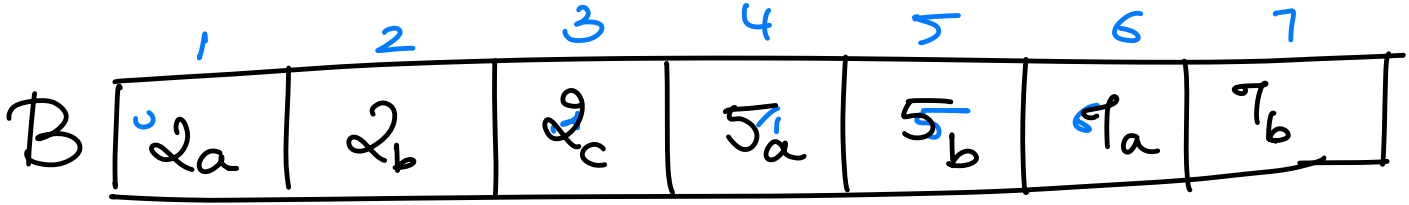




(where to put the numbers is given by above array)

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Step 4: (Getting the sorted output).



Input:

	1	2	3	4	5	6	7	8
A	2_a	1_a	1_b	5_a	7_a	5_b	7_b	1_c

$k = 7$

Step 1 & 2:

	0	1	2	3	4	5	6	7
C	0	3	1	0	0	2	0	2

Step 3:

	0	1	2	3	4	5	6	7
C	0	3 → 2	4 → 3	4	4	6 → 5 → 4	6	8 → 6

Step 4:

	1	2	3	4	5	6	7	8
B	1_a	1_b	1_c	2_a	5_a	5_b	7_a	7_b

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Note:

Order of repeated numbers in the input is maintained in the output by Counting sort.

Such type of sort is called Stable Sort.

Passes = 3 = No of Columns.

